

Parallel Analysis

2026-06-06

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# =====
# Comprehensive R Script for Horn's Parallel Analysis and Velicer's MAP Test
# Project: COVID-19 Human Mobility Factor Analysis (Sri Lanka Survey Dataset)
# Dataset: 40 Mobility and Behavioral Adaptation Items (N = 1,898)
# =====

# -----
# Step 1: Install and Load Required Libraries
# -----
# The 'psych' package contains optimized functions for EFA/PCA diagnostics,
# including fa.parallel() and MAP().
if (!requireNamespace("psych", quietly = TRUE)) {
  install.packages("psych")
}
library(psych)

# -----
# Step 2: Load and Prepare the Dataset
# -----
# Update the data_path to the location of your CSV file on your local machine.
# The uploaded file contains 40 items with a '_PP' suffix.
data_path <- "C:/Users/ihpof/OneDrive/Documents/AddressingReviewsMobility/Factor Analysis Review/Parallel Analysis and MAP test/dataset_for_unsupervised_ML.csv"

if (!file.exists(data_path)) {
  stop("Dataset not found! Please place the CSV in your active working directory or update 'data_path'.")
}

# Read raw dataset
mobility_data <- read.csv(data_path, header = TRUE, stringsAsFactors = FALSE)

# Verify dimensions and column headers
cat("--- Dataset Diagnostics ---\n")
```

```
## --- Dataset Diagnostics ---
```

```
cat("Observations (Rows):", nrow(mobility_data), "\n") # Expected: 1898
```

```
## Observations (Rows): 1898
```

```
cat("Variables (Columns):", ncol(mobility_data), "\n\n") # Expected: 40
```

```
## Variables (Columns): 40
```

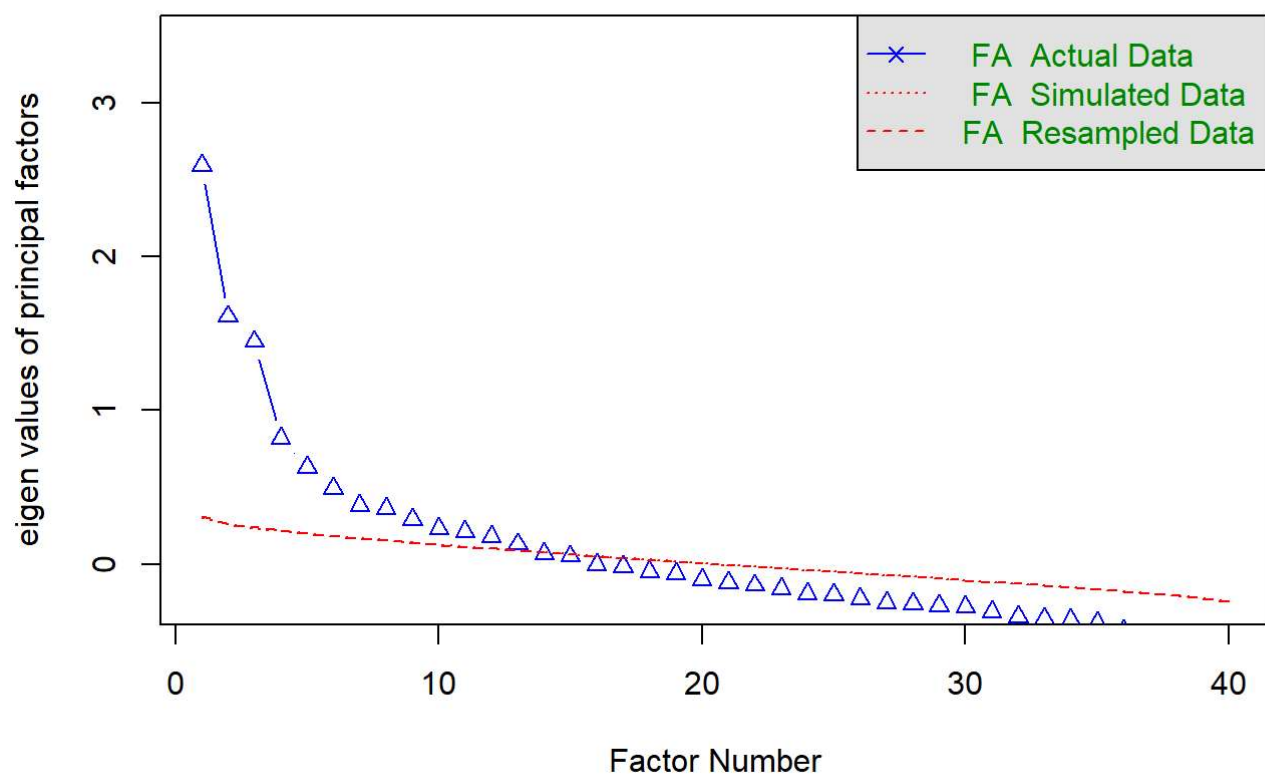
```
print(head(mobility_data[, 1:5])) # Print a preview of the first 5 variables
```

```
##   sof_neighborhood_shop_PP sof_main_super_market_PP sof_mini_super_market_PP
## 1           3.000000           0.000000           0.000000
## 2           1.333333           0.000000           0.000000
## 3           0.000000           0.666667           0.666667
## 4          -1.000000           1.333333           0.000000
## 5           0.000000           0.000000           0.000000
## 6          -1.666667           0.000000           0.000000
##   sof_home_delivery_or_mobile_shops_PP sof_own_field_PP
## 1                                   0                0
## 2                                   0                0
## 3                                  -1                0
## 4                                  -1                0
## 5                                   0                0
## 6                                  -1               -1
```

```
# -----
# Step 3: Horn's Parallel Analysis (PA)
# -----
# Horn's PA contrasts empirical eigenvalues against eigenvalues computed from
# random data matrices of identical dimensions (1,898 x 40).[2, 3]
# We use 'fa = "both"' to plot and compare eigenvalues for both Principal
# Components (PCA) and Exploratory Factor Analysis (EFA/Principal Axis).
# Set seed for reproducible Monte-Carlo simulations.
set.seed(12345)

# Running the analysis
parallel_results <- fa.parallel(
  x = mobility_data,
  fa = "fa",
  fm = "minres",
  n.iter = 500,
  error.bars = FALSE,
  main = "Parallel Analysis for Latent Structure (40 Raw Items)"
)
```

Parallel Analysis for Latent Structure (40 Raw Items)



```
## Parallel analysis suggests that the number of factors = 13 and the number of components = NA
```

```
# Printing only the number of factors
cat("Suggested number of factors:", parallel_results$nfact, "\n")
```

```
## Suggested number of factors: 13
```